Cost Analysis

2026-06-15

Contents

1 Assumptions

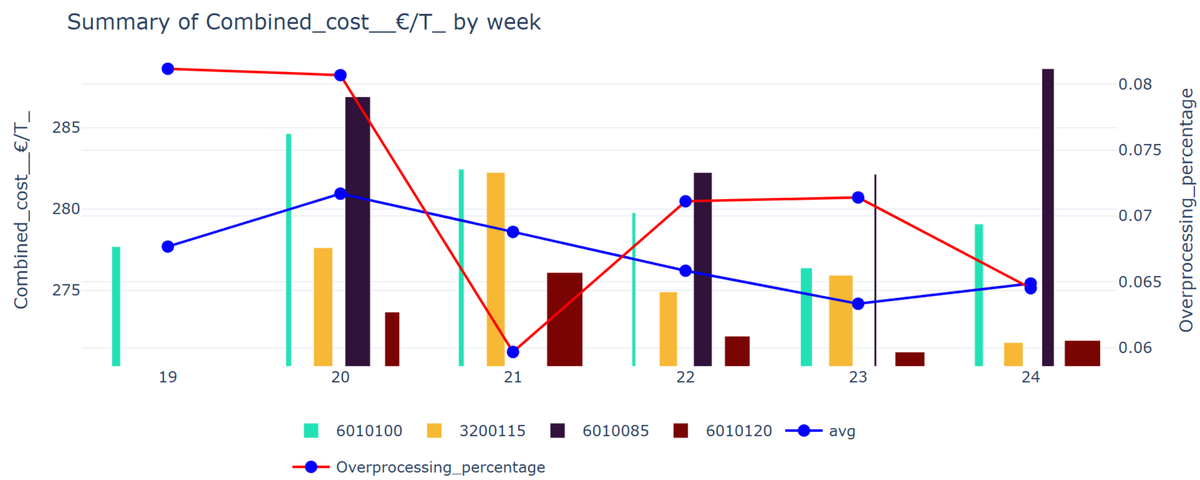
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

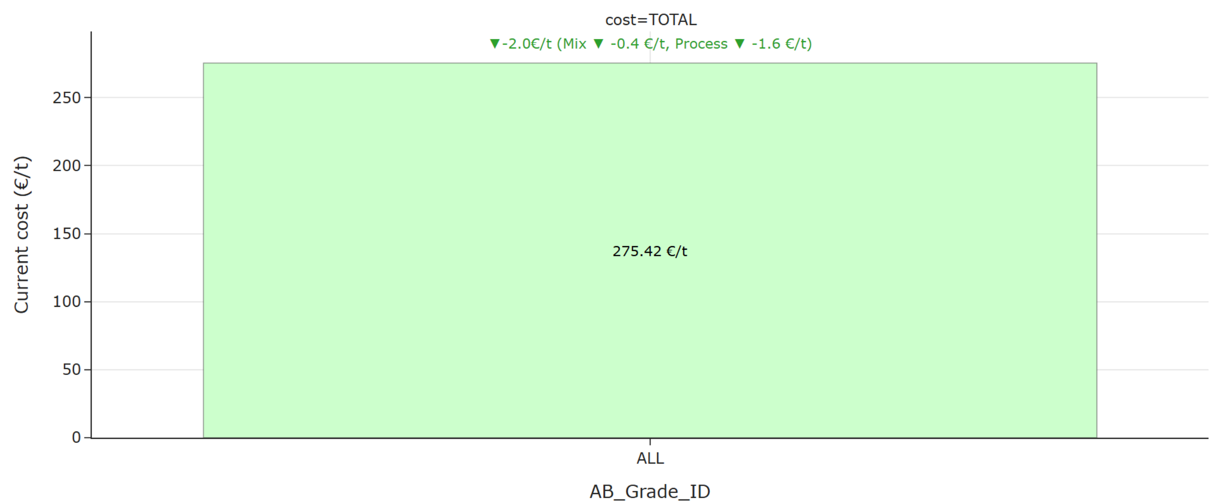
- **Period Analyzed**: May 10, 2026 – June 15, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.
  - **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
  - **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

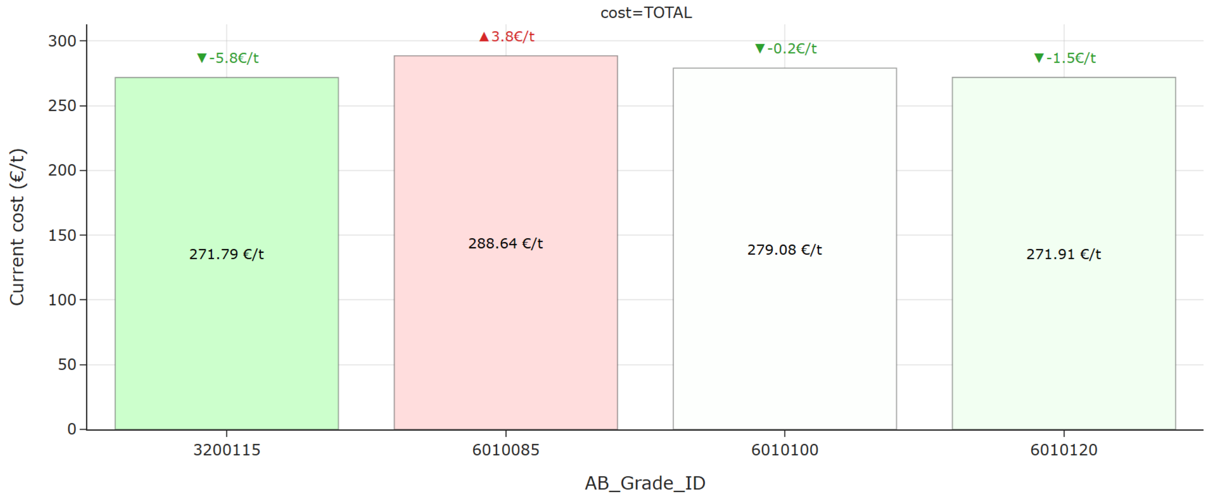
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw a decrease of 2.01 €/t, resulting in 275.42 €/t.

Breaking down the overall cost change, -0.42 €/t was driven by a cheaper mix of grades, and -1.58 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 5.79 €/t, resulting in 271.79 €/t.

For grade 6010085, TOTAL cost saw an increase of 3.84 €/t, resulting in 288.64 €/t.

For grade 6010100, TOTAL cost saw a decrease of 0.24 €/t, resulting in 279.08 €/t.

For grade 6010120, TOTAL cost saw a decrease of 1.53 €/t, resulting in 271.91 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010085 (+3.84 €/t) and the strongest cost decrease was observed for grade 3200115 (-5.79 €/t). These movements stand out and may require further investigation.

Average change by component:

- **Steam__€/T** decreased on average (-0.91 €/t).
- **Starch__€/T** increased on average (+0.37 €/t).
- **Fibre_cost__€/T** decreased on average (-0.26 €/t).
- **Electricity__€/T** decreased on average (-0.13 €/t).
- **Chemicals__€/T** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Starch__€/T** (+0.37 €/t on average), with the highest increase in grade 6010085 (+1.73 €/t) and the largest average decrease comes from **Steam__€/T** (-0.91 €/t on average), with the biggest drop in grade 3200115 (-4.62 €/t). These components stand out and may require further investigation.

Average change by steam:

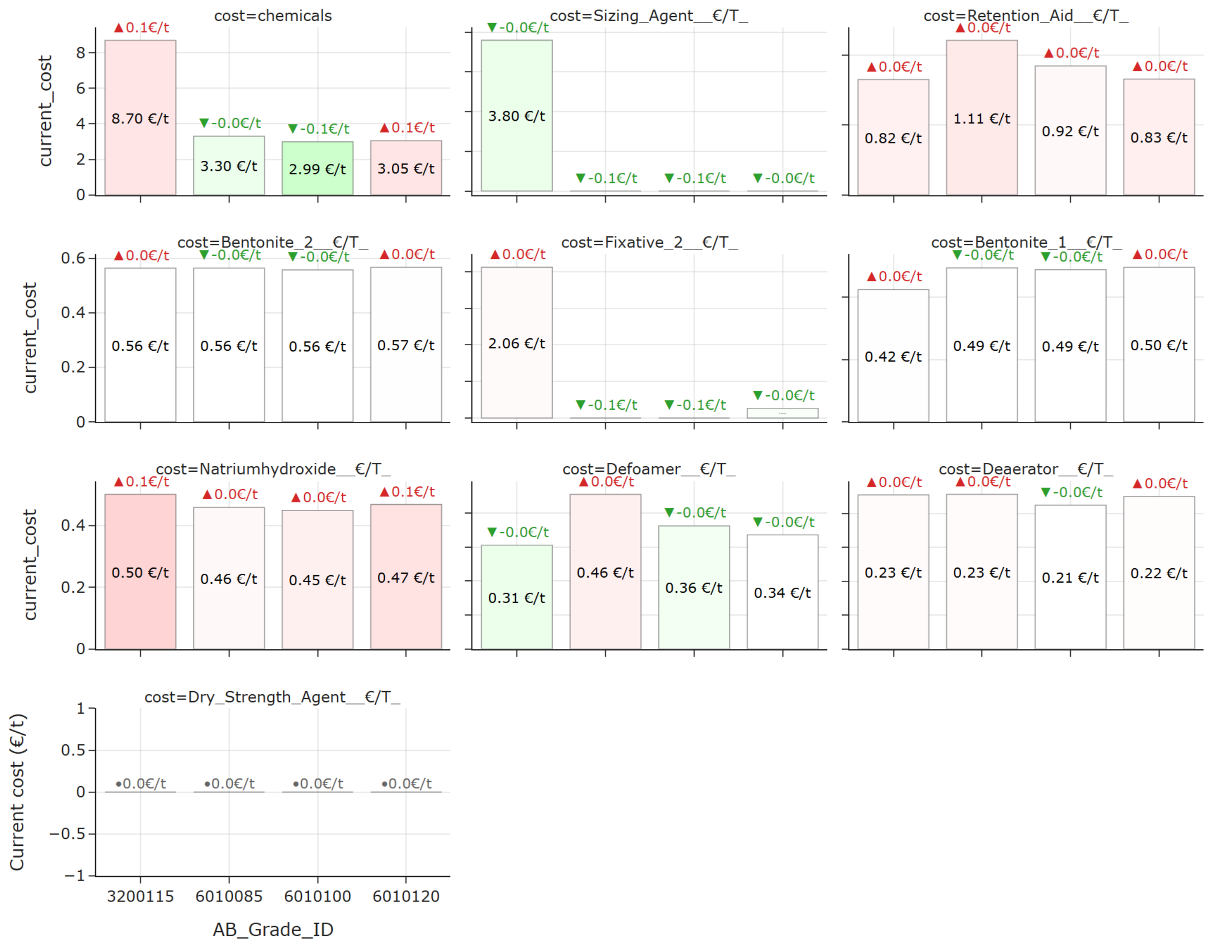
- **Steam_Predryers** decreased on average (-0.71 €/t).
- **Steam_Afterdryers** decreased on average (-0.23 €/t).
- **Steam_Whitewater** increased on average (+0.12 €/t).
- **Steam_Heatexchangers** decreased on average (-0.05 €/t).
- **Steam_Starchkitchen** decreased on average (-0.03 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Whitewater** (+0.12 €/t on average), with the highest increase in grade 6010100 (+0.71 €/t) and the largest average decrease comes from **Steam_Predryers** (-0.71 €/t on average), with the biggest drop in grade 3200115 (-1.75 €/t). These components stand out and may require further investigation.

Average change by electricity:

- **Electricity_Wiresection** decreased on average (-0.10 €/t).
- **Electricity_Specific** increased on average (+0.08 €/t).
- **Electricity_Afterdryer** decreased on average (-0.07 €/t).
- **Electricity_Other** decreased on average (-0.03 €/t).
- **Electricity_Predryer** decreased on average (-0.02 €/t).
- **Electricity_Freqconverters** increased on average (+0.02 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Specific** (+0.08 €/t on average), with the highest increase in grade 6010100 (+0.20 €/t) and the largest average decrease comes from **Electricity_Wiresection** (-0.10 €/t on average), with the biggest drop in grade 6010100 (-0.15 €/t). These components stand out and may require further investigation.

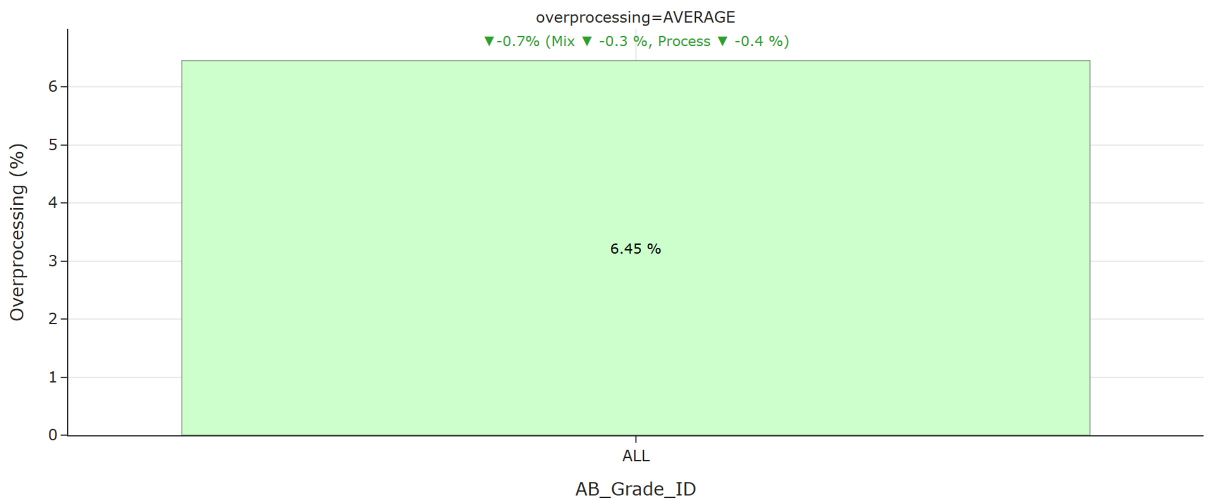


Average change by chemicals:

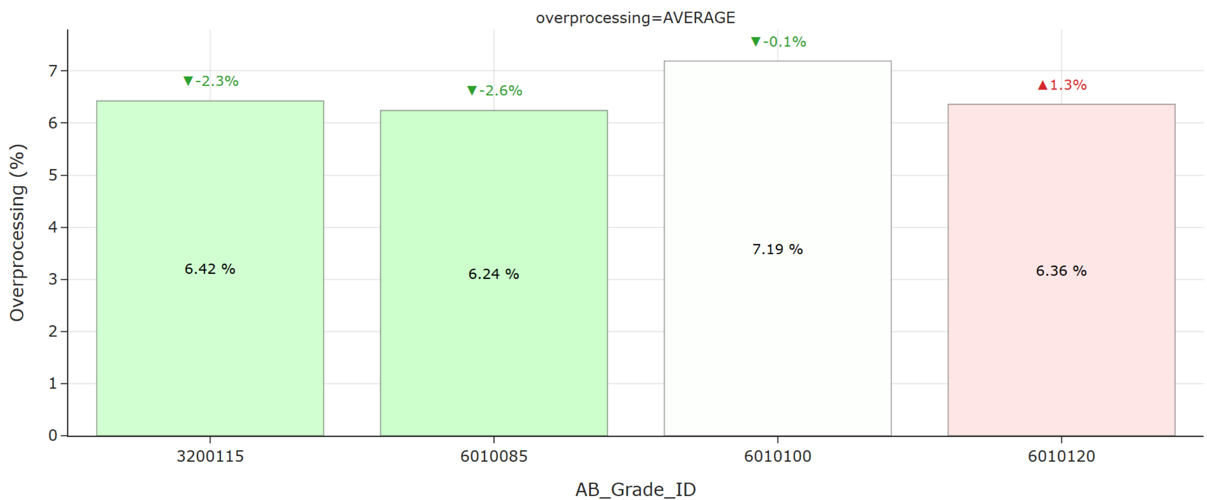
- **Natriumhydroxide__€/T** increased on average (+0.06 €/t).
- **Sizing_Agent__€/T** decreased on average (-0.05 €/t).
- **Retention_Aid__€/T** increased on average (+0.03 €/t).
- **Defoamer__€/T** decreased on average (-0.01 €/t).
- **Deaerator__€/T** increased on average (+0.01 €/t).
- **Bentonite_2__€/T** decreased on average (-0.00 €/t).
- **Fixative_2__€/T** decreased on average (-0.00 €/t).
- **Bentonite_1__€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Natriumhydroxide__€/T** (+0.06 €/t on average), with the highest increase in grade 3200115

(+0.10 €/t) and the largest average decrease comes from **Sizing__Agent__€/T** (-0.05 €/t on average), with the biggest drop in grade 6010085 (-0.06 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 0.68 %, resulting in 6.45 %.



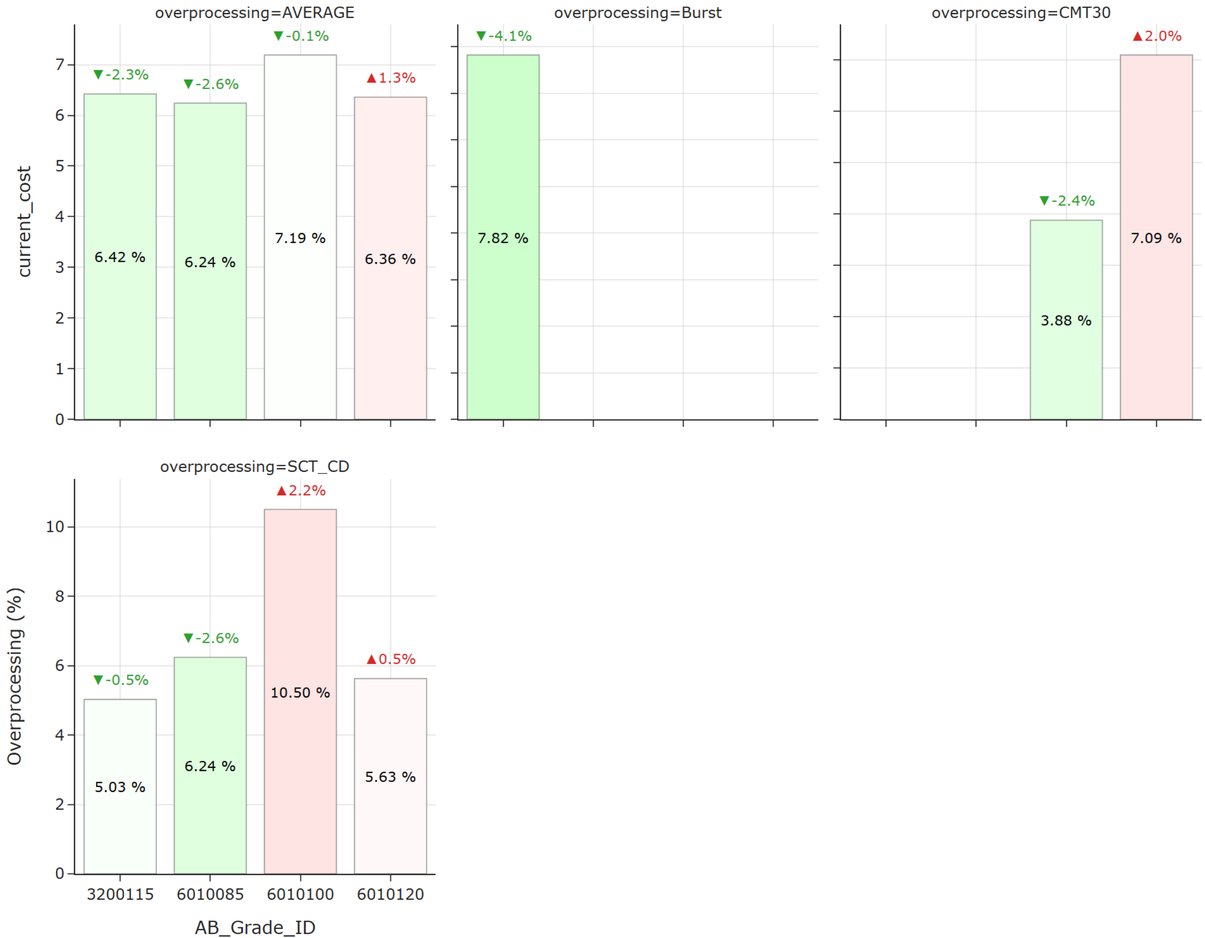
For grade 3200115, TOTAL overprocessing saw a decrease of 2.32 %, resulting in 6.42 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 2.61 %, resulting in 6.24 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 0.13 %, resulting in 7.19 %.

For grade 6010120, TOTAL overprocessing saw an increase of 1.27 %, resulting in 6.36 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010120 (+1.27 %) and the strongest overprocessing decrease was observed for grade 6010085 (-2.61 %). These movements stand out and may require further investigation.



Average change by component:

- **Burst** decreased on average (-4.12 %).
- **CMT30** decreased on average (-0.20 %).
- **SCT_CD** decreased on average (-0.11 %).

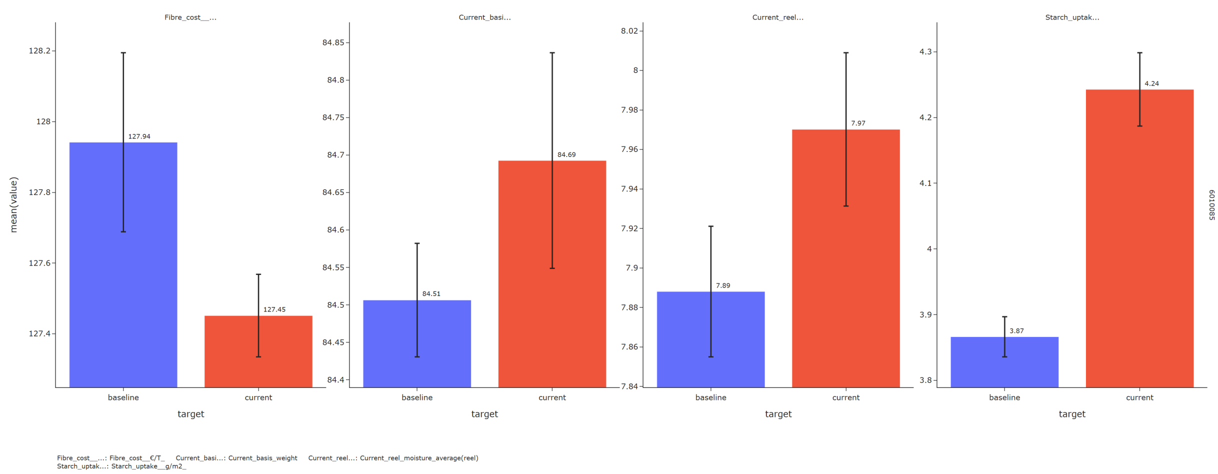
A more detailed breakdown by component shows that the largest average decrease comes from **Burst** (-4.12 % on average), with the biggest drop in grade 3200115 (-4.12 %). These components stand out and may require further investigation.

3 Breakdown by grades

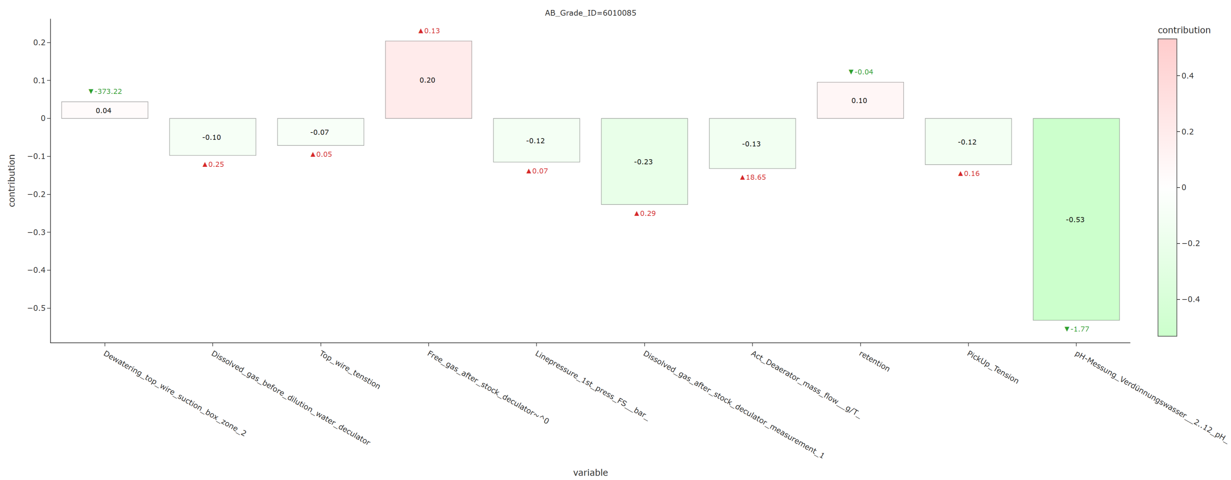
3.1 6010085

3.1.1 Fiber cost

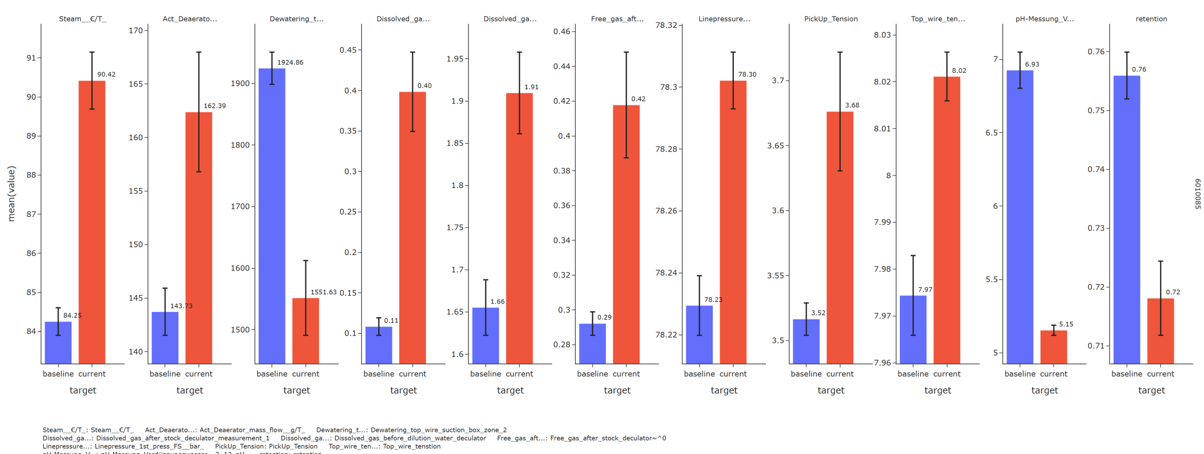
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Starch uptake g/m2 increased (+0.38)



3.1.2 Steam cost

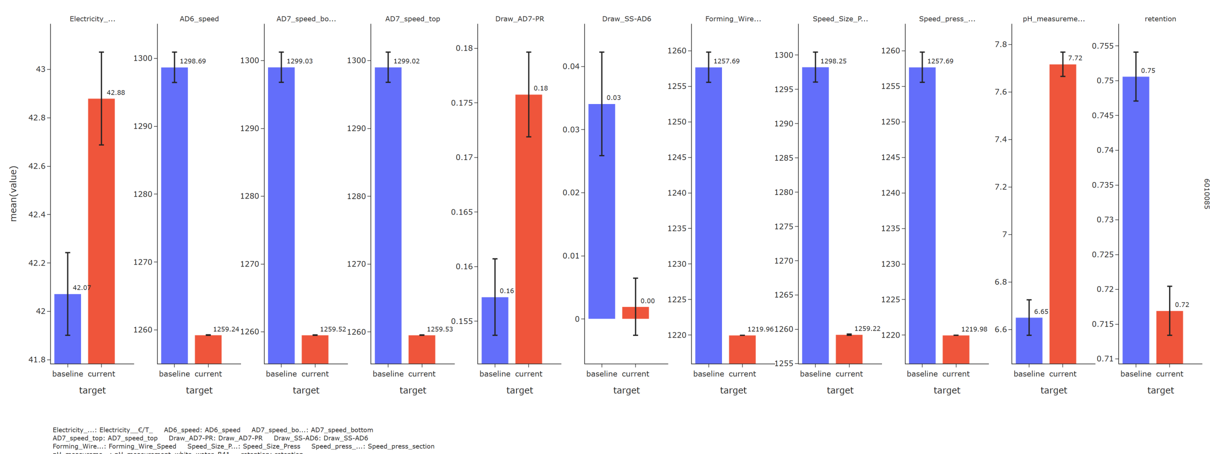


For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.77) - Dissolved gas after stock deculator measurement 1 increased (+0.29) - Act Deaerator mass flow g/T increased (+18.65) - PickUp Tension increased (+0.16) - Linepressure 1st press FS bar increased (+0.07) - Dissolved gas before dilution water deculator increased (+0.25) **Cost-increasing drivers**: - Free gas after stock deculator~^0 increased (+0.13)

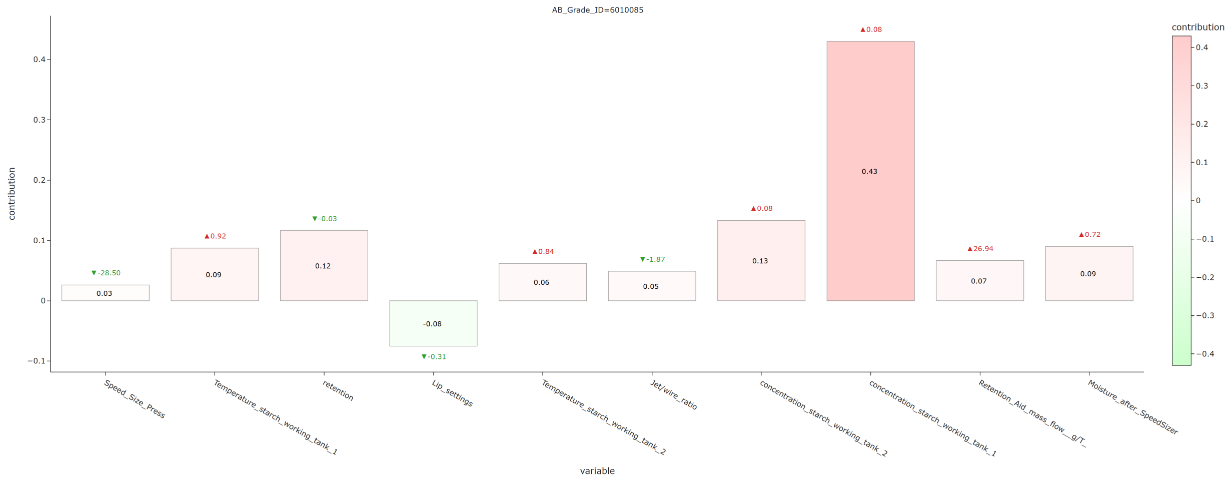


3.1.3 Electricity cost

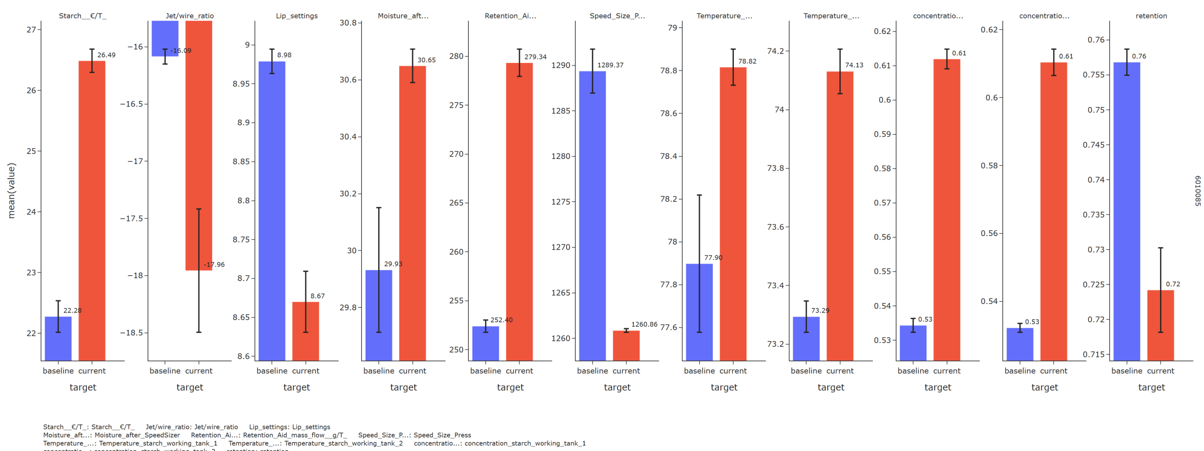
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Draw SS-AD6 decreased (-0.03) **Cost-increasing drivers**: - Retention decreased (-0.03) - PH measurement white water B41 increased (+1.07) - Draw AD7-PR increased (+0.02) - Forming Wire Speed decreased (-37.72)



3.1.4 Starch cost

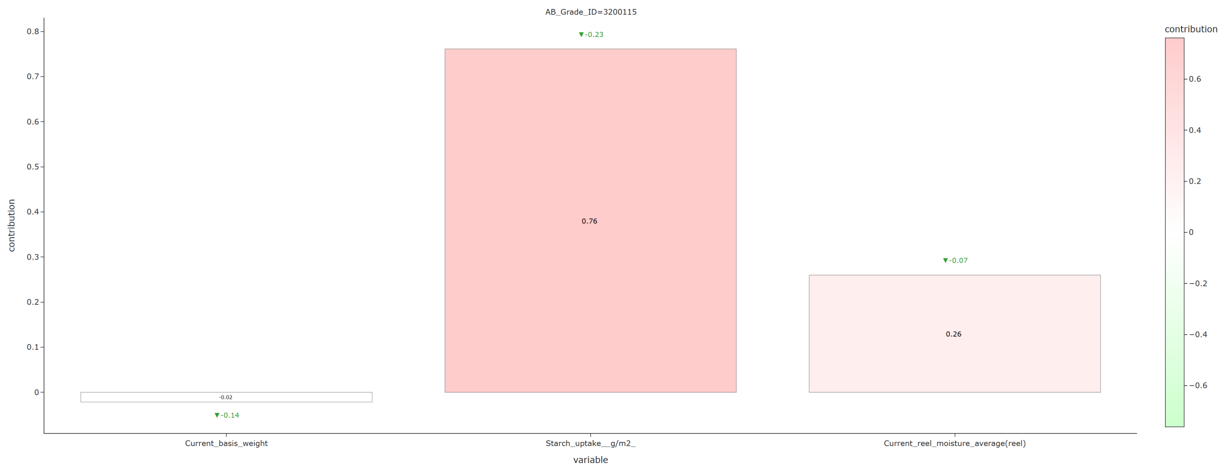


For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Concentration starch working tank 1 increased (+0.08) - Concentration starch working tank 2 increased (+0.08) - Retention decreased (-0.03)

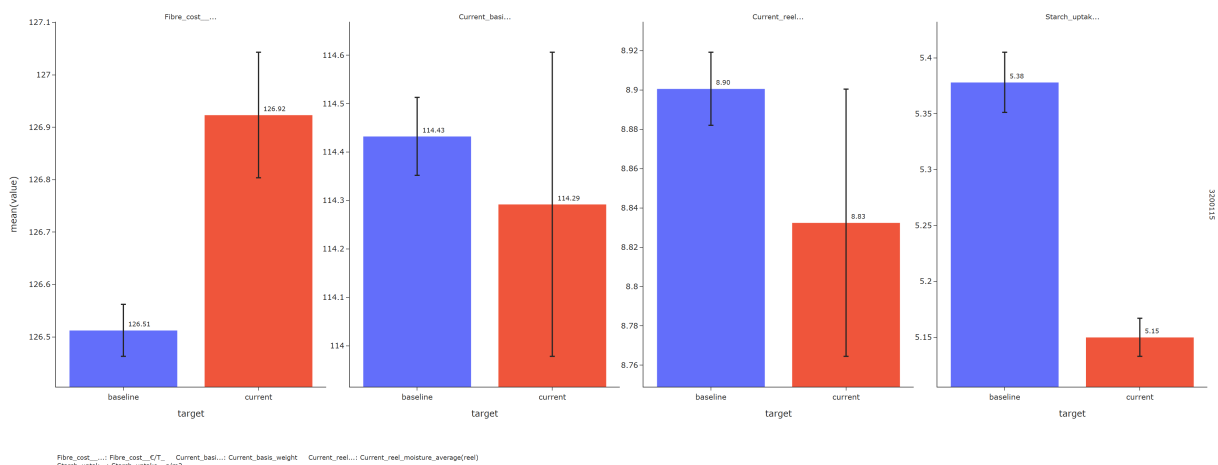


3.2 3200115

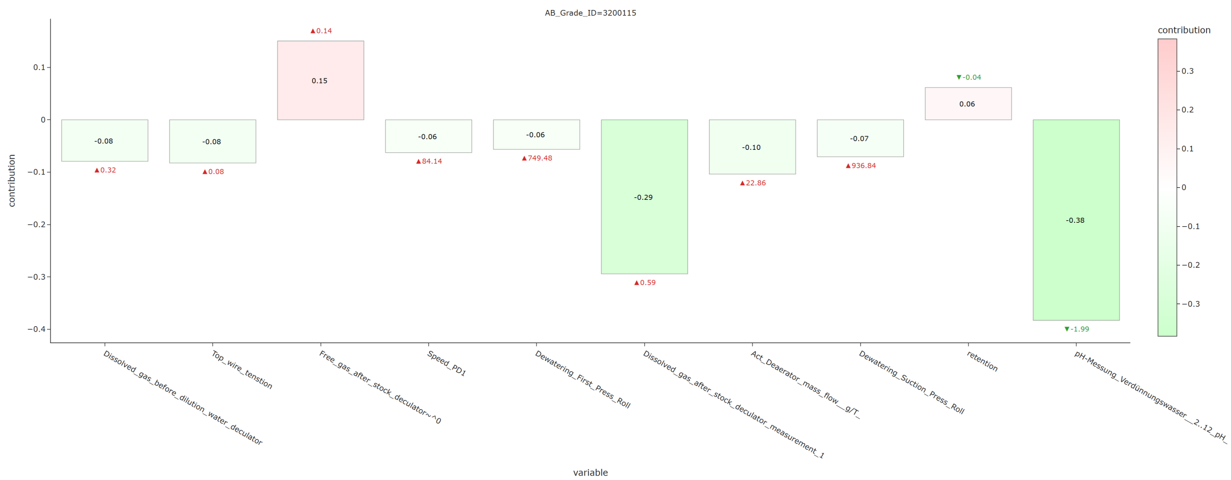
3.2.1 Fiber cost



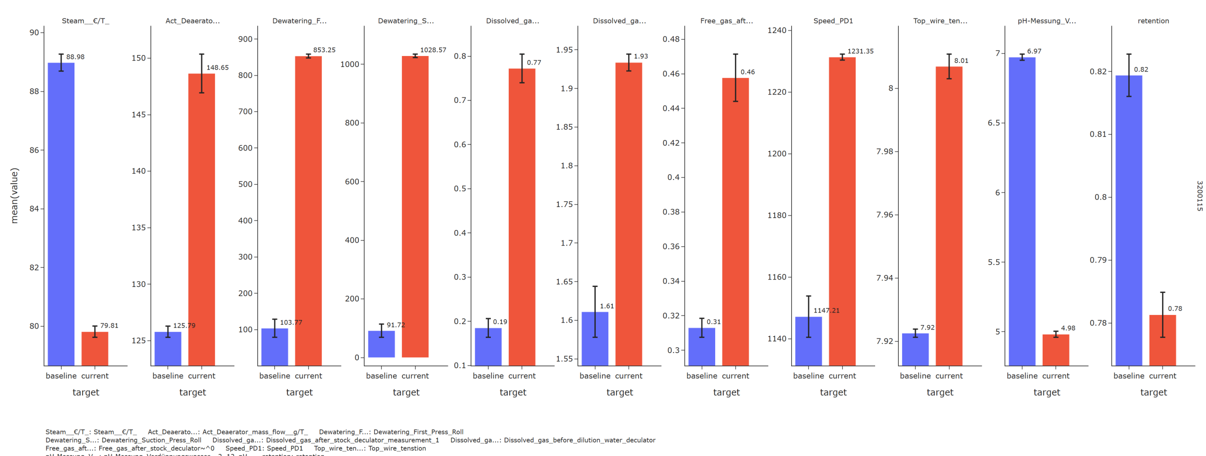
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Starch uptake g/m2 decreased (-0.23)



3.2.2 Steam cost

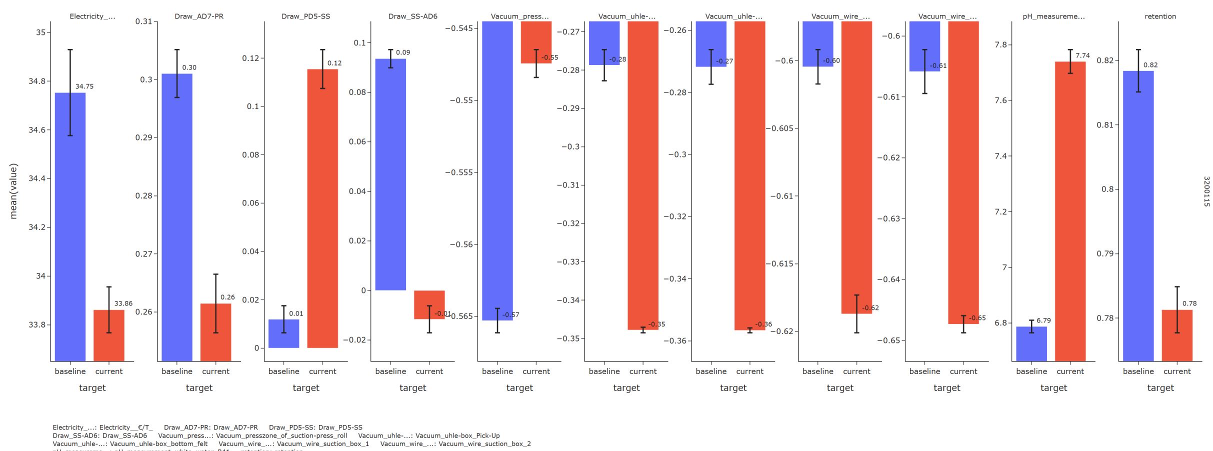


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.99) - Dissolved gas after stock deculator measurement 1 increased (+0.59) - Act Deaerator mass flow g/T increased (+22.86) - Top wire tenstion increased (+0.08) - Dissolved gas before dilution water deculator increased (+0.32) - Dewatering Suction Press Roll increased (+936.84) **Cost-increasing drivers**: - Free gas after stock deculator~^0 increased (+0.14)

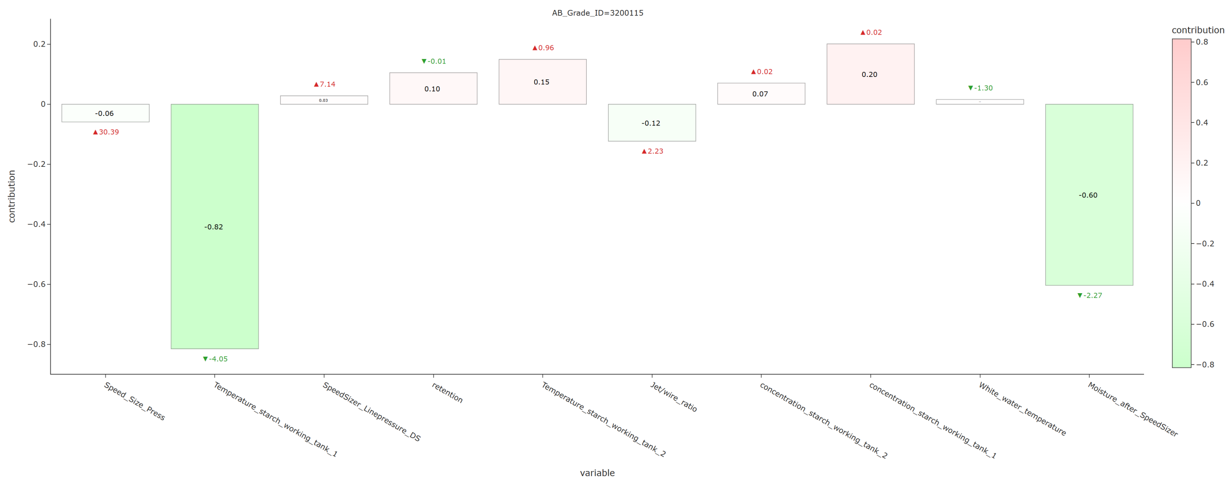


3.2.3 Electricity cost

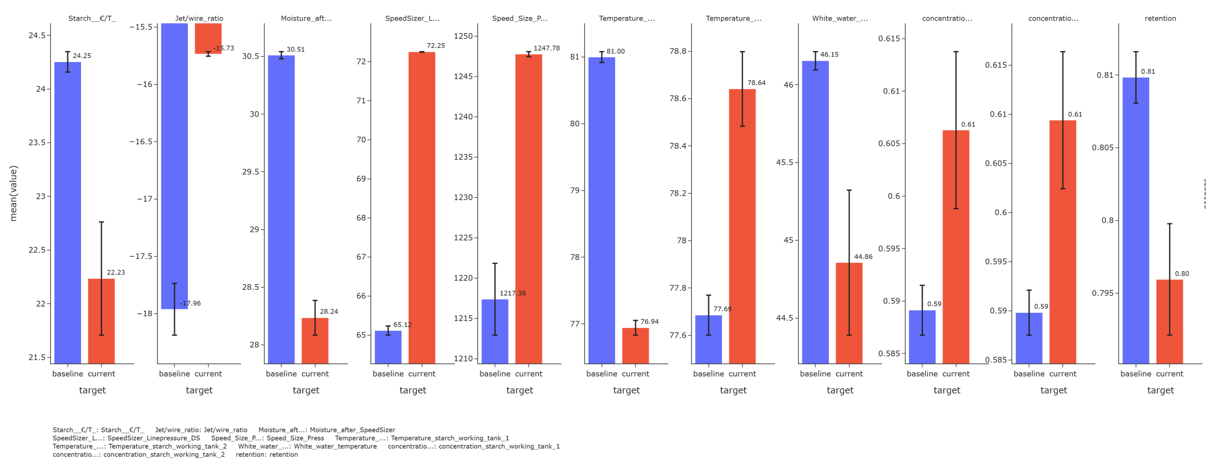
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum wire suction box 1 decreased (-0.02) - Vacuum uhle-box Pick-Up decreased (-0.07) - Vacuum presszone of suction-press roll increased (+0.02) **Cost-increasing drivers**: - Retention decreased (-0.04) - Vacuum wire suction box 2 decreased (-0.04) - Vacuum uhle-box bottom felt decreased (-0.08)



3.2.4 Starch cost

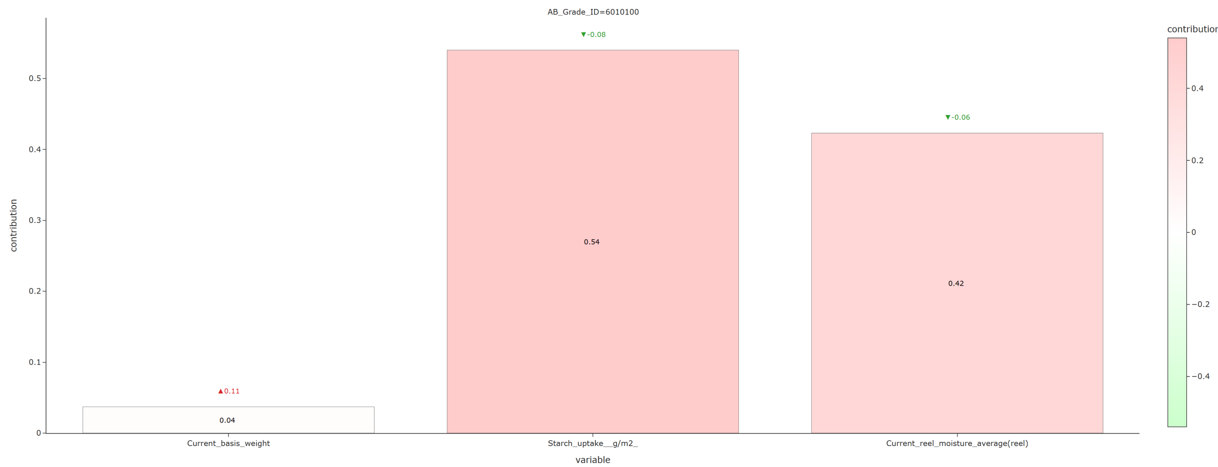


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Temperature starch working tank 1 decreased (-4.05) - Moisture after SpeedSizer decreased (-2.27) **Cost-increasing drivers**: - Concentration starch working tank 1 increased (+0.02)

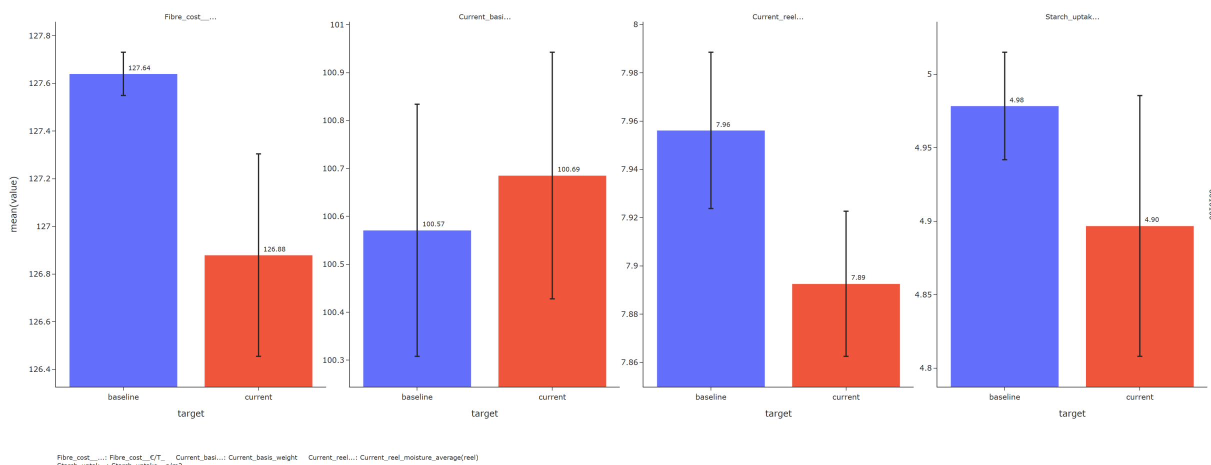


3.3 6010100

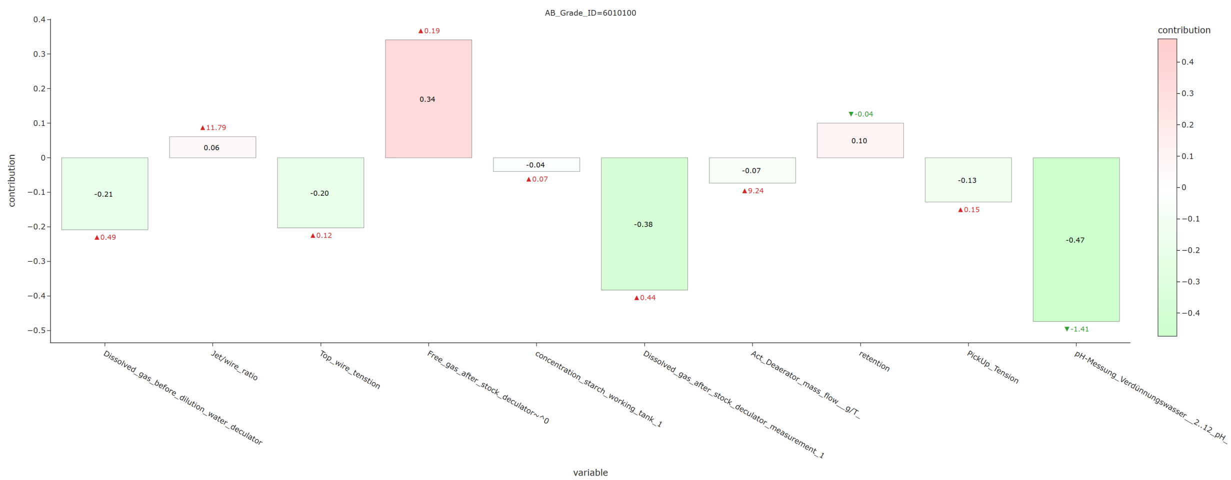
3.3.1 Fiber cost



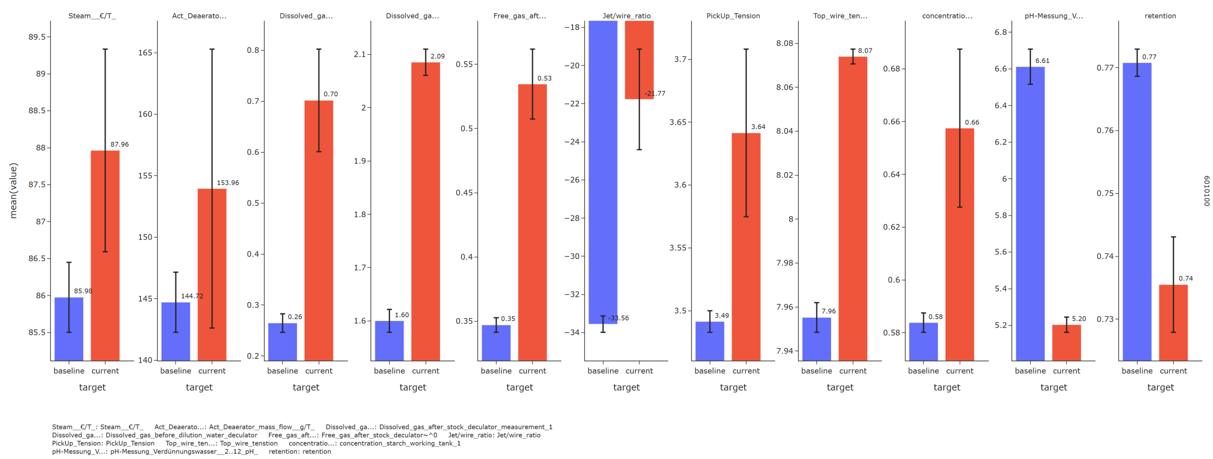
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Starch uptake g/m2 decreased (-0.08)



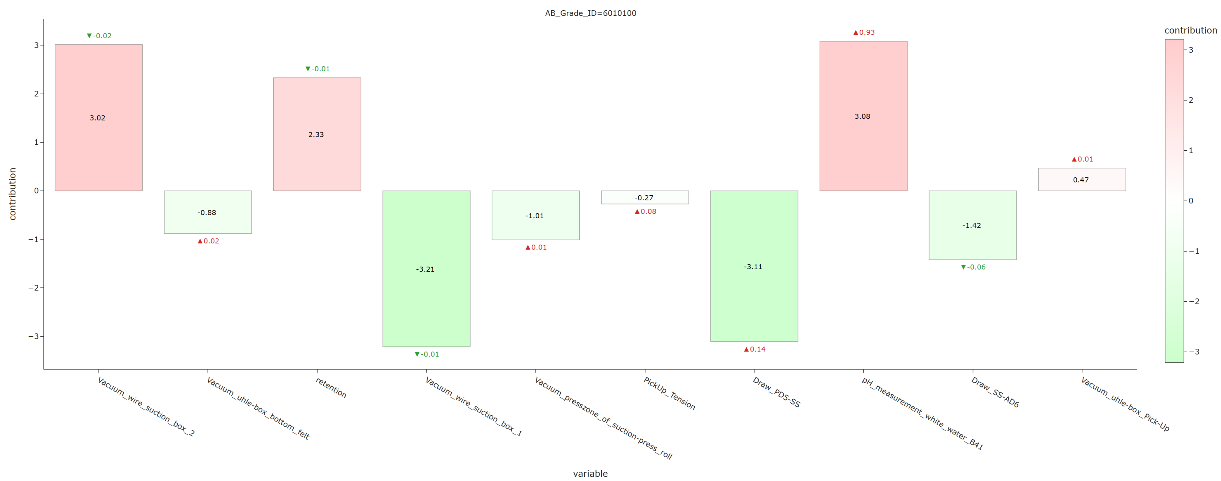
3.3.2 Steam cost



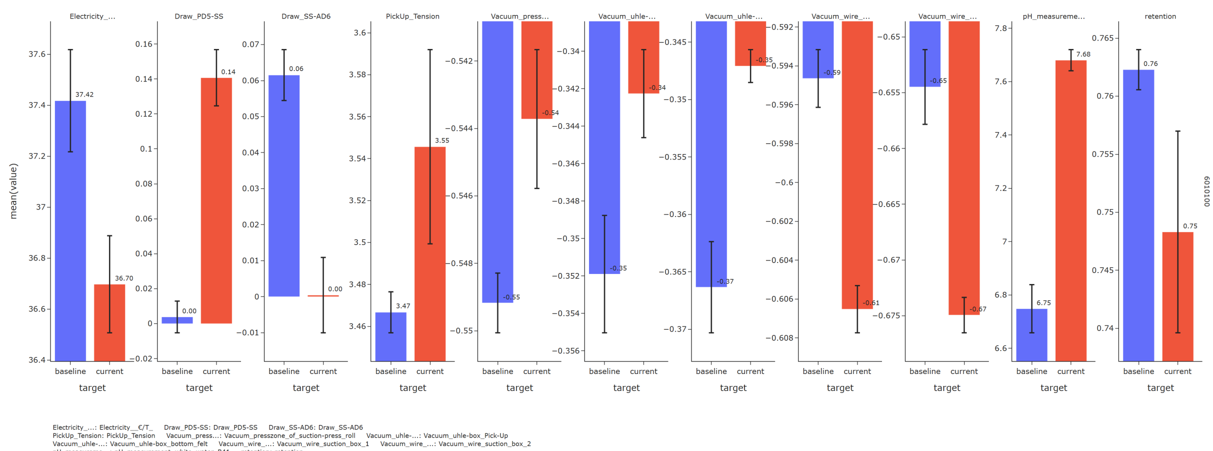
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.41) - Dissolved gas after stock deculator measurement 1 increased (+0.44) - Dissolved gas before dilution water deculator increased (+0.49) - Top wire tenstion increased (+0.12) - PickUp Tension increased (+0.15) **Cost-increasing drivers**: - Free gas after stock deculator~^0 increased (+0.19)



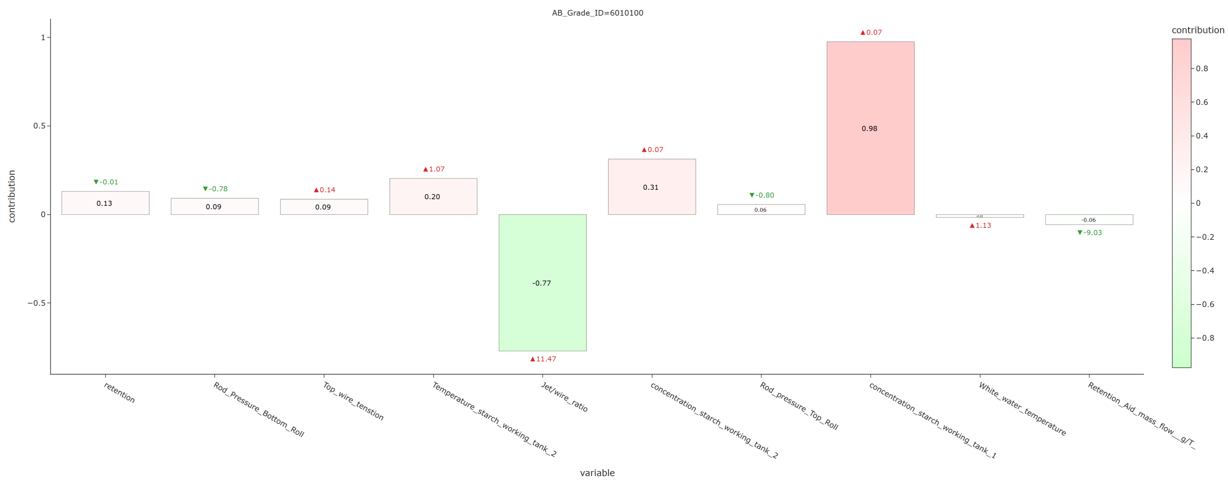
3.3.3 Electricity cost



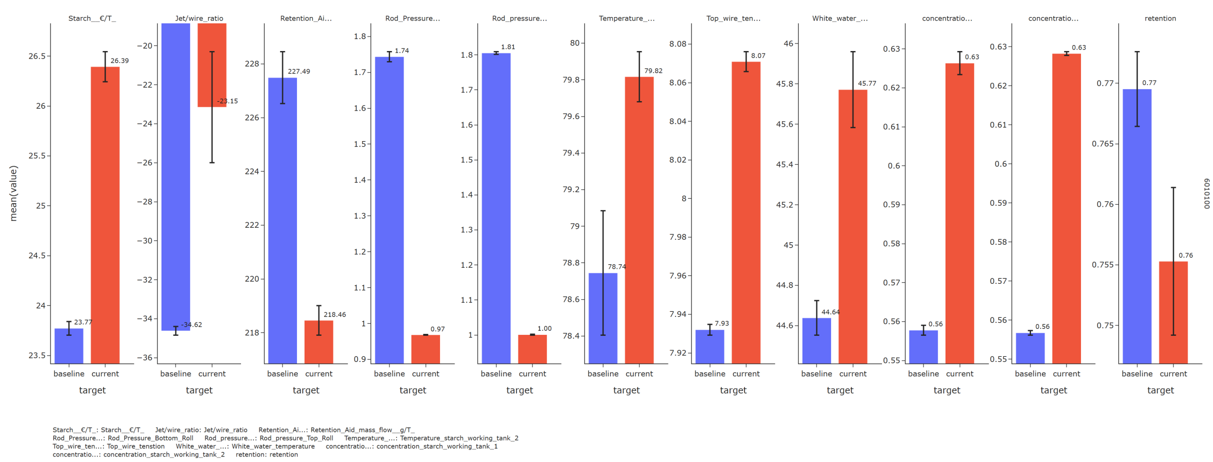
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum wire suction box 1 decreased (-0.01) - Draw PD5-SS increased (+0.14)



3.3.4 Starch cost

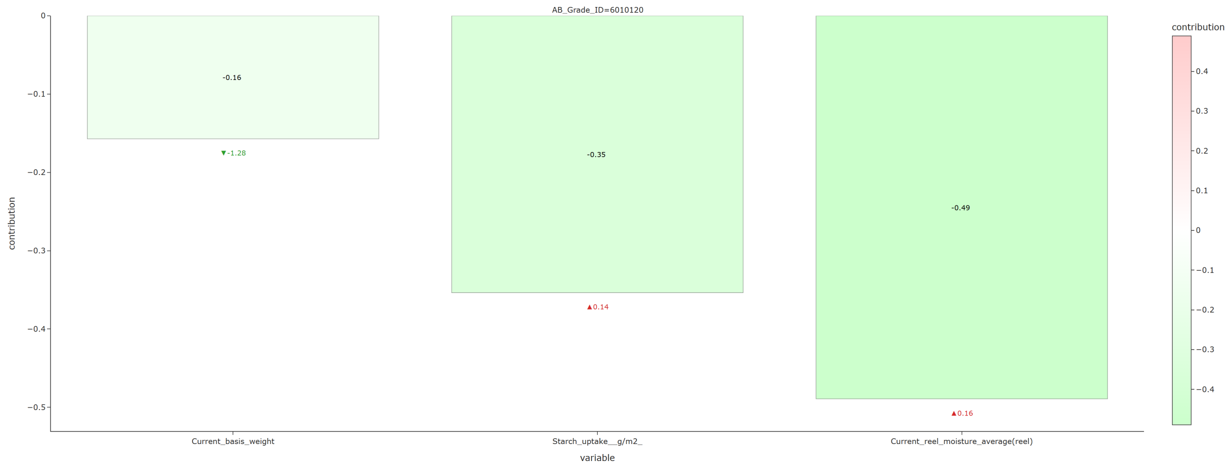


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Jet/wire ratio increased (+11.47) **Cost-increasing drivers**: - Concentration starch working tank 1 increased (+0.07) - Concentration starch working tank 2 increased (+0.07)

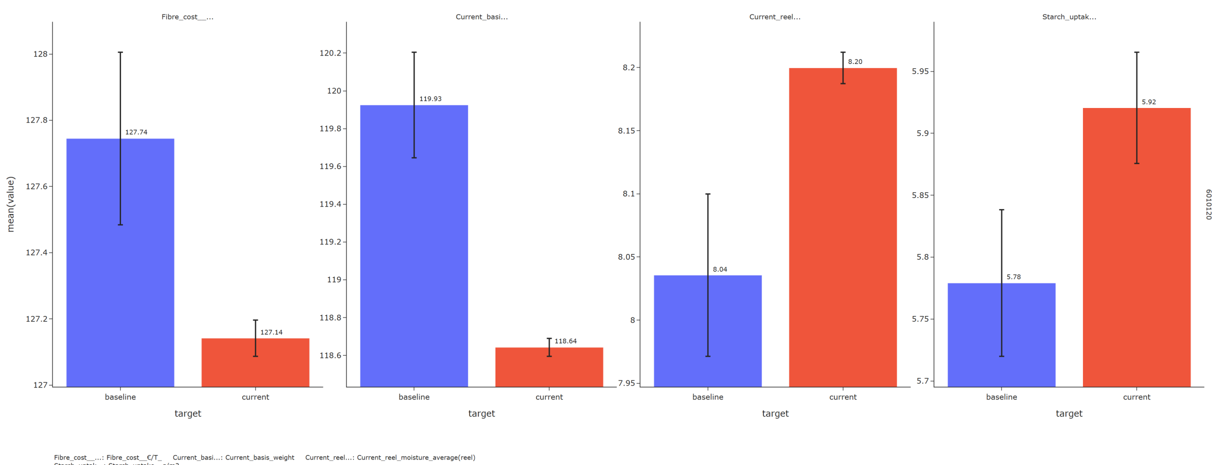


3.4 6010120

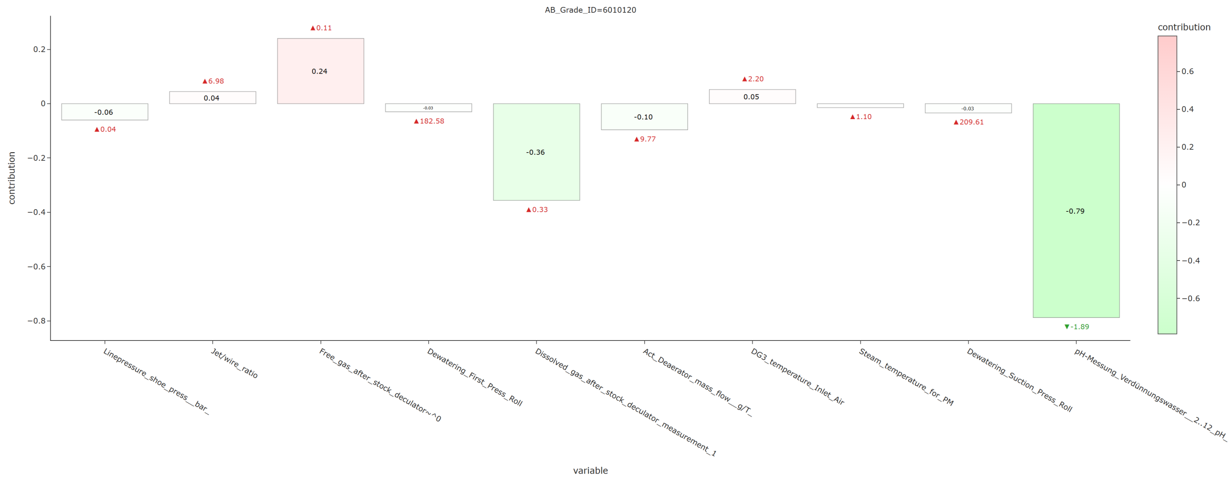
3.4.1 Fiber cost



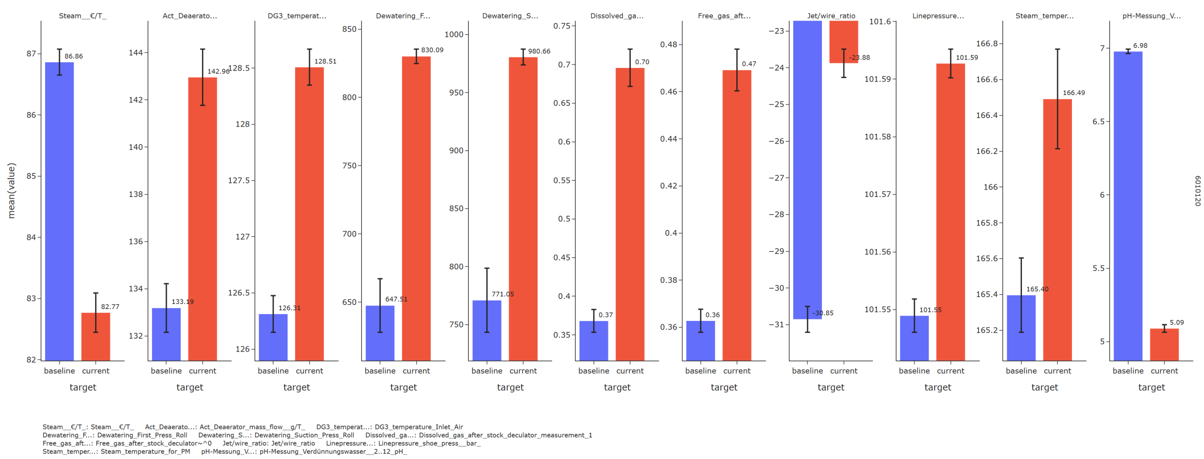
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Current reel moisture average(reel) increased (+0.16)



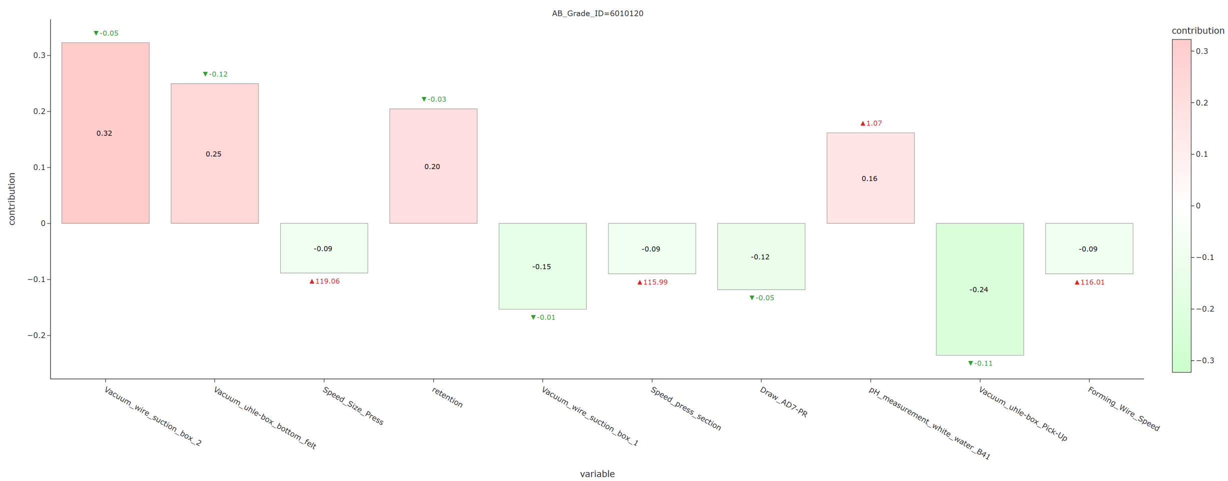
3.4.2 Steam cost



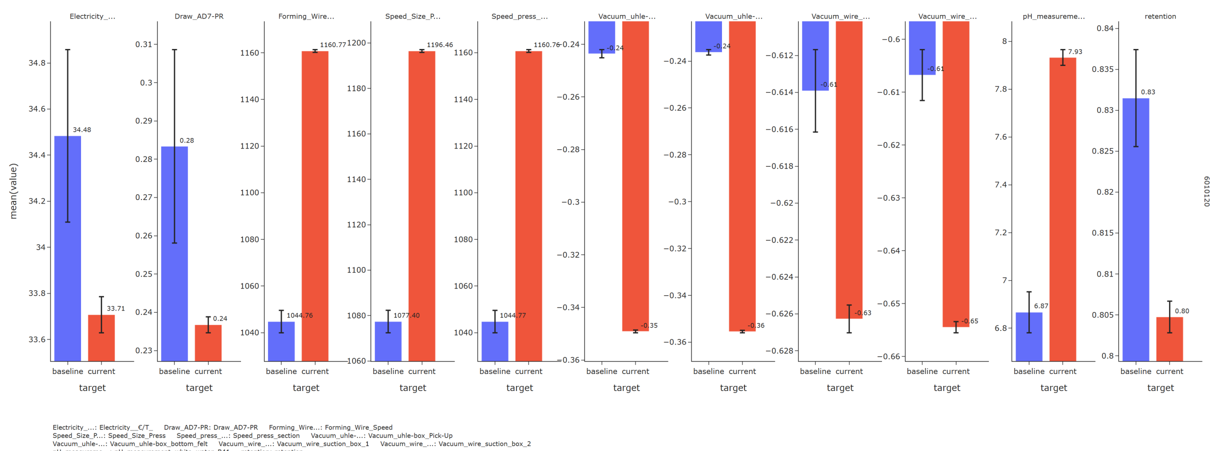
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.89) - Dissolved gas after stock deculator measurement 1 increased (+0.33) - Act Deaerator mass flow g/T increased (+9.77) **Cost-increasing drivers**: - Free gas after stock deculator~^0 increased (+0.11)



3.4.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Vacuum uhle-box Pick-Up decreased (-0.11) - Vacuum wire suction box 1 decreased (-0.01) **Cost-increasing drivers**: - Vacuum wire suction box 2 decreased (-0.05) - Vacuum uhle-box bottom felt decreased (-0.12) - Retention decreased (-0.03) - PH measurement white water B41 increased (+1.07)



3.4.4 Starch cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Moisture after SpeedSizer decreased (-1.72) - Temperature starch working tank 2 decreased (-1.38) - Lip settings decreased (-0.40)